

A Foundational Framework for Information-Physical Interconversion

Author: Yue Lu

Version: 1.6 (Axiom-Purified & Quantitatively Mapped Edition)

Three Core Principles

I. Statistical Foundation (Mapping-Layer Causal Constraint Principle)

All physical foundations originate from information statistics. Physical constants (e.g., $\hbar$ and c) are not intrinsic properties of matter. Instead, they are the **emergent ontological ultraviolet boundary** and the **emergent causal upper bound of propagation** arising from the mapping of information-space onto physical reality. Therefore, information theory is not merely a descriptive tool; it serves as the fundamental metalanguage for analyzing the underlying laws of physics.

II. Residual Observation (Instantiation-Boundary Principle)

All physical observations are essentially processes of information transmission and logical residual evaluation. The Planck scale marks the boundary for logical instantiation within the system. Observation acts as a dynamic mechanism that drives the evolution of physical systems through continuous evaluation and compensation of data discrepancies.

III. Functional Interchangeability (Digital-Analog Conversion Principle)

Physical theories and information theories are functionally equivalent and interchangeable. This duality establishes a real-world **"digital-analog conversion toolkit"**: by defining well-defined rigid boundaries across macroscopic, mesoscopic, and microscopic scales, gaps within physical models can be remedied via algorithmic optimization, while intricate informational structures can be instantiated and mapped following physical principles.

Conclusion

SRE-Dynamics delivers an innovative two-way tool for scientific research. By establishing the equivalence between the informational and physical domains, we gain the capability to resolve previously intractable scientific paradoxes through **cross-domain recoding**. By optimizing the system's logical pathways within the constraint scope of underlying causal-instantiation boundaries, we can achieve precise intervention upon observable physical outcomes.